Hasan Shaikh

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Work Experience

Quantitative Imaging Research and Artificial Intelligence Lab (QIRAIL) Tamil Nadu, India
Data Scientist, Christian Medical College Vellore, Dept. of Radiation Oncology *Aug. 2024 – Present*

[P1] Large-Scale Prospective Imaging Study – DBT/Wellcome Trust India Alliance

- Built end-to-end DICOM retrieval pipeline via Orthanc PACS and Python for a prospective radiomics cohort of **2,000+ head and neck cancer patients**.
- Automated tumour segmentation and PyRadiomics feature extraction across the full cohort, standardising inputs for downstream prognostic modelling.
- Deployed AWS S3 backup pipelines with automated error recovery, ensuring longitudinal data integrity across a five-year multisite radiotherapy study.
- Coordinated annotation workflows across radiation oncology sites, resolving DICOM metadata inconsistencies and validating clinical record completeness at scale.
- Gained end-to-end experience in the clinical data lifecycle of a **DBT/Wellcome Trust India Alliance**-funded prospective oncology imaging study.

[P2] Radiomics-Based Risk Stratification in Head and Neck Cancer

- Engineered a **10-feature radiomics-based prognostic signature** (2 clinical + 8 radiomic) for locoregional recurrence prediction in head and neck cancer.
- Benchmarked **5 metaheuristic optimisation algorithms** (Genetic Algorithm etc.) across 1,000+ candidate features to identify the most stable, generalisable subset.
- Achieved **AUC 0.82 (95% CI: 0.62–0.95)** across 327 primary-radiation cohort; applied SHAP for per-prediction clinical interpretability.
- Recognised as a **Poster Highlight at ESTRO 2026**, Stockholm, Sweden – the flagship European radiotherapy research conference.

[P3] Automated Segmentation of Head and Neck Cancer Using 3D CNNs – Collaboration with NIT Surathkal

- Curated a **multi-institutional CT dataset** of 167 head and neck cancer cases (137 MAASTRO + 30 CMC Vellore) for cross-cohort segmentation evaluation.
- Trained and validated a **3D nnU-Net** for CT-only gross tumour volume segmentation, achieving Dice 0.62 (MAASTRO), 0.63 (CMC), and 0.65 (combined).
- Co-authored **AGSR-Net** – anatomy-guided skip-connection recalibration improving Dice (0.70→0.73) and HD95 (33.92→21.22 mm) across 896 patients; under submission at *International Journal of Biomedical Imaging*.

[P4] CHAVI: CompreHensive Digital ArchiVe of Cancer Imaging – India's First National Oncology Biobank

- Contributed **304+ de-identified cases** with validated clinical and imaging metadata to CHAVI – India's first national oncology imaging biobank.
- Processed and anonymised DICOM datasets using XNAT and Orthanc, ensuring compliance with national data governance and multisite interoperability standards.
- Collaborated with **Tata Memorial Centre** and **IIT Kharagpur** to harmonise imaging data across institutions for pan-India cancer AI research.

[P5] Reproducibility Study: CNN-Based Head and Neck Cancer Prognosis

- Systematically audited a peer-reviewed **CNN-based head and neck cancer prognosis model** against **CLAIM** reporting frameworks.
- Identified **data leakage** – preprocessing statistics computed on the full dataset, including the test set – inflating all reported performance metrics across three clinical outcomes.
- Corrected the full pipeline, reproduced results from scratch, and obtained **formal written acknowledgement** from the original authors.
- Submitted as *"Can reproducibility be achieved?"* to **Scientific Reports**, Nature Portfolio [Under Submission]; code available at .
- Demonstrated that published clinical AI benchmarks can be systematically unreliable – advancing reproducibility standards in radiation oncology AI.

STARlab Capital — Lucknow, India
Research Analyst — *Dec. 2023 – June 2024*

- Designed and backtested **volatility-based options strategies** (Nebula, ARUT, A2) using OptionNet Explorer, Mesosim, and OptiTrade.
- Led development contributions to OptiTrade's open-source backtesting repository, improving trade log systems and internal pipeline reliability through structured software engineering practices.

Publications & Research

[1] Asjad Nabeel P, Hannah Mary Thomas T, **Hasan Shaikh**, Sathya A, Rajendra Benny Kuchipudi, Jino Wilson Victor, Simon P Pavamani, Balu Krishna Sasidharan, Jeny Rajan, *"Anatomy-Guided Skip-connection Recalibration for CT-based Head and Neck Tumor Auto-segmentation"*, *International Journal of Biomedical Imaging*. **[Under Submission]**

[2] **Hasan Shaikh**, Sathya A, Praveenraj C, Balu Krishna S, Aparna Irodi, Rajendra Benny Kuchipudi, Jino Wilson Victor, Manu Mathew, Rajesh Isiah, Simon Pavamani, Joy Mammen, Hannah Mary Thomas T, *"Can reproducibility be achieved? External validation of 'Reproducible' image-based prognostic models for head and neck cancer"*, *Scientific Reports, Nature Portfolio*. **[Under Submission]**

[3] Balu Krishna S, Amal Joseph Varghese, **Hasan Shaikh** et al., *"Development and validation of a prospective radiomics-clinical signature for locoregional recurrence in patients with locally advanced head and neck cancer"*, *European Society for Radiotherapy and Oncology (ESTRO)*, Stockholm, Sweden, 2026. [Abstract] [Poster]

[4] **Hasan Shaikh**, Balu Krishna S, Amal Joseph Varghese, et al., *"Metaheuristic-Driven Machine Learning Pipelines for Radiomics-Based Prediction of Locoregional Recurrence in Head and Neck Cancer"*, In: *Proceedings of the International Conference on Artificial Intelligence for Healthcare (AIHC), Lecture Notes in Electrical Engineering, Springer Nature*, 2025. **[Accepted]**

[5] Piyus Prabhanjans, Asjad Nabeel P, Aparna V K, Hannah Mary Thomas T, Balu Krishna Sasidharan, **Hasan Shaikh**, Amal Joseph Varghese, Rajendra Benny Kuchipudi, Simon Pavamani, Jeny Rajan, *"Automated Segmentation of Head and Neck Cancer from CT Images Using 3D Convolutional Neural Networks"*, *medRxiv* (preprint), 2026. https://doi.org/10.64898/2026.03.12.26347996

[6] **Hasan Shaikh** and Rashid Ali, *"Cancer Survival Prediction Using Artificial Intelligence: Current Status and Future Prospects"*, *Data Science in the Medical Field, Academic Press, Elsevier*, 2024. https://doi.org/10.1016/B978-0-443-24028-7.00016-7 **[Book Chapter]**

[7] **Hasan Shaikh**, Amal Joseph Varghese, Balu Krishna S, et al., *"Before We Treat, Can We Tell? A Locoregional Recurrence Signature in Head & Neck"*, *15th Research Day*, Christian Medical College, Vellore, India, 2025. **[Poster]**

[8] **Hasan Shaikh**, Amal Joseph Varghese, Hannah Mary Thomas T et al., *"Can CT Radiomics Predict Recurrence in Head and Neck Cancer? Early Results from a Prospective Imaging Trial"*, *14th Research Day*, Christian Medical College, Vellore, India, 2024. **[Poster]**

Master's Dissertation — Multimodal Deep Learning for Cancer Survival Prediction

Hasan Shaikh, *"Multimodal Data Analytics for Predicting the Survival of Cancer Patients"*, M.Tech Thesis, Department of Computer Engineering, Aligarh Muslim University, 2023. Advisor: Prof. Rashid Ali. **Published as:** *"Cancer Survival Prediction Using Artificial Intelligence: Current Status and Future Prospects"*, in *Data Science in the Medical Field*, Academic Press, Elsevier, 2024.

- Designed **MGAttCNNMD** (Multimodal Gated Attention Convolutional Neural Network integrating Multidimensional Data) – three parallel Gated Attention CNNs (clinical, CNV, gene expression) fused via stacked Random Forest for breast cancer survival prediction.
- Benchmarked against all single-modality baselines on **METABRIC** dataset, achieving **91.2% survival prediction accuracy**. Demonstrated that multi-omics fusion (clinical + genomic + transcriptomic) yields superior prognostic discrimination over any unimodal model.

Technical Skills

Machine Learning & DL: TensorFlow, Keras, scikit-learn, PyRadiomics, nnU-Net
Programming Language : Python (NumPy, Pandas, SciPy), Bash
Tools : 3D Slicer, Orthanc PACS, XNAT, DICOM, NIfTI, Docker, Github
Specialized Methods : Radiomics, Metaheuristic Techniques, Graph Neural Networks

Education

Master of Technology in Computer Science & Engineering Nov. 2021 – Nov. 2023
Aligarh Muslim University (AMU), CGPA: 8.80 / 10.00 *Aligarh, India*

Bachelor of Technology in Computer Science and Engineering Aug. 2017 – Jul. 2021
Dr. A.P.J. Abdul Kalam Technical University (AKTU), CGPA: 8.04 / 10.00 *Lucknow, India*

Academic Achievements & Honors

- **Third Prize, Oral Presentation** at the 2nd National Symposium on Health Data and AI (March 2025) — Presented *"Automated Segmentation of Head and Neck Cancer from CT Images Using 3D nnU-Net"*, awarded by the BioMedical Informatics Unit, CMC Vellore, India.
- **Honors Degree** — Ranked in the top 1% out of 128 students in the undergraduate engineering program.
- **NPTEL Courses conducted by IITs (Elite + Silver Certified):**
  - Data Analytics with Python (80%) (Link)
  - Essential Mathematics for Machine Learning
  - Deep Learning
  - Python for Data Science (78%) (Link)
  - Machine Learning
  - Demystifying the Brain (Link)

Relevant Courses

Relevant courses completed during my Bachelor's and Master's programs include:

- Artificial Intelligence
- Machine Learning
- Web Mining & Searching
- Big Data Analytics
- Image Processing & Computer Vision
- Application of Soft Computing
- Data Structures
- Applied Linear Algebra
- Design & Analysis of Algorithms
- Database Management Systems
- Data Warehousing & Data Mining
- Distributed Systems

Conferences & Workshops

1. Presented paper at the **International Conference on Artificial Intelligence for Healthcare (AIHC)**, organized by National Institute of Technology (NIT) Calicut, India (December 10 - 12, 2025) on Metaheuristic-driven radiomics-based locoregional recurrence prediction in head and neck cancer. [*Lecture Notes in Electrical Engineering, Springer*] (Link)

2. Organized the **1st & 2nd Workshop on Radiomics and Auto Segmentation**, conducted by the Department of Radiation Oncology & Quantitative Imaging Research and Artificial Intelligence Lab, Christian Medical College (CMC) Vellore, Tamil Nadu, India (November 14–15, 2025 & March 27-28, 2026). (Link)

3. Organized the **Head and Neck Cancer Survivorship Program**, conducted by the Department of Radiation Oncology, Christian Medical College (CMC) Vellore, Tamil Nadu, India (September 12–13, 2025). Themes: "Together We Overcome, Together We Heal" and "De-Escalation Strategies in Head and Neck Cancer."

4. Participated and served as part of the Organizing Team for the **2nd Annual Winter Symposium on Health Data and AI**, conducted by Biomedical Informatics Unit, Christian Medical College (CMC) Vellore, Tamil Nadu, India (March 13–15, 2025).

5. Attended the Continuing Medical Education (CME) program on **Revolution and Precision in Radiation Oncology**, Ida B. Scudder Cancer Center, Christian Medical College (CMC) Vellore, Tamil Nadu, India (March 1, 2025).

6. Participated in the **14th & 15th Annual Research Day**, organized by the Office of Research, Christian Medical College (CMC) Vellore, Tamil Nadu, India (October 24–25, 2024 & 2025).

7. **AI & ML for Engineering & Social Sciences Research**, 2023 IEEE Computational Intelligence Society (CIS) Summer School, organized by Malaviya National Institute of Technology Jaipur (MNIT) Jaipur, India, 4 – 8 Sep. 2023. (Link)

8. **7th Summer School on AI with Focus on Computer Vision & ML**, coordinated by International Institute of Information Technology (IIIT) Hyderabad, India, 1 – 31 Aug. 2023. (Link)

9. **Emerging Research Trends in Computational Intelligence Techniques to Address Challenges in Biomedical Data and Imaging**, 2022 IEEE CIS Summer School, organized by National Institute of Technology (NIT) Arunachal Pradesh, India, 7 – 11 Nov. 2022. (Link)

Referees

Dr. Hannah Mary Thomas T
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